INCH-POUND

MIL-I-45914A
25 April 1991
SUPERSEDING
MIL-I-45914
1 December 1967

MILITARY SPECIFICATION

INSERT, SCREW THREAD - LOCKED IN, KEY LOCKED, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers locked in, screw thread inserts which have integrated locking keys to prevent rotation of the insert when installed in the tapped holes.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5340

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SPECIFICATIONS

FEDERAL

QQ-A-225/6	-Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn or Cold Finished, 2024
QQ-A-225/8	-Aluminum Alloy Bar, Rod, Wire, Rolled, Drawn, or Cold Finished, 6061
QQ-A-250/4	-Aluminum Alloy 2024, Plate and Sheet
QQ-A-250/11	-Aluminum Alloy 6061, Plate and Sheet
QQ-P-35	-Passivation Treatments for Austenitic, Ferretic, and Martensitic Corrosion-Resisting Steel (Fastening Devices)
QQ-P-416	-Plating, Cadmium (Electrodeposited)
PPP-H-1581	-Hardware (Fasteners and Related Items), Packaging of

MILITARY

MIL-S-5626	-Steel, Chrome-Molybdenum (4140) Bars and Reforging Stock (For Aircraft Applications)
MIL-H-6875	-Heat Treatment of Steels (Aircraft Practice), Process for
MIL-S-8879	-Screw Threads, Controlled Radius Root and Increased Minor Diameter, General Specification for
MIL-I-17214	-Indicator, Permeability; Low MU (Go-No Go)
MIL-L-46010	-Lubricant, Solid Film, Heat Cured

STANDARDS

FEDERAL

FED-STD-H28/2	-Screw-Thread Standards for Federal Services, Section 2, Unified Inch Screw Threads-UN and UNR Thread Forms.
FED-STD-H28/20	-Screw-Thread Standards for Federal Services, Section 20 Inspection Methods for Acceptability of UN, UNR, UNJ, M and MJ Screw-Threads.

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MIL-STD-105	-Sampling Procedures and tables for Inspection by Attributes
MIL-STD-1312	-Fasteners, Test Method
MIL-STD-1949	-Inspection process - magnetic particle
MIL-STD-6866	-Inspection, penetrant method of
MS51830	-Insert, Screw Thread-Locked in, Key Locked, Miniature and Lightweight
MS51831	-Insert, Screw Thread-Locked In, Key Locked, Heavy Duty
MS51832	-Insert, Screw Thread-Locked In, Key Locked, Extra Heavy Duty
MS51835	-Inserts and Studs, Locked In, Key Locked, Hole, Dimensions for and Assembly of

HANDBOOKS

MILITARY

MIL-HDBK-57	-Listing of Fastener Manufacturer's Identification Symbols.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AEROSPACE MATERIAL SPECIFICATION (AMS)

AMS 5640 Steel Bars, Forgings, Corrosion Resistant
AMS 5737 Steel Bars, Forgings, and Tubing, Corrosion and Heat Resistant
AMS 6322 Steel Bars, Forgings and Rings

(Application for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form)

ANSI/ASME B46.1 - Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036)

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ASTM

ASTM A 580 -Stainless and Heat-Resisting Steel Wire

ASTM A 582 -Free Machining Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold Finished

(Application for copies should be addressed to ASTM, 1916 Race Sreeet, Philadelphia, Pennsylvania 19103-1187.)

(Non-government standards and other publications are normally available from the organizations that prepare or that distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained in which case the exception will be identified in the text and cited in the solicitation.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable. Inserts and keys shall be fabricated of the following materials, as specified (see 3.1).

3.2.1 Alloy steel. Alloy steel shall be types 4140 (UNS G41400) in accordance with MIL-S-5626 or 8740 (UNS G87400) in accordance with AMS 6322.

3.2.2 Corrosion-resistant steel. Corrosion-resistant steel inserts shall be types 303 (UNS S30300) or 303SE (UNS S30323) in accordance with AMS 5640, ASTM A 582 or type A286 conforming to AMS 5737 (UNS S66286). Type A286 corrosion resistant steel inserts shall have magnetic permeability of 2.0 maximum (Air = 1.0) for a field of H=200 oersteds.

3.2.3 Locking keys. Insert locking keys shall be made from corrosion-resistant steel, type 302, chemical composition of ASTM A 580 only.

3.3 Heat treatment. Inserts made of alloy steel in accordance with 3.2.1 shall be heat treated in accordance with MIL-H-6875.

3.4 Cadmium plating and surface treatment. Inserts shall be furnished with a cadmium plating or surface treatment as specified herein and 3.1.

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3.4.1 Cadmium plating. Inserts fabricated from alloy steel shall be cadmium plated in accordance with QQ-P-416, Type II, Class 3, except stress durability test is not required. The entire insert shall be cadmium plated. The locking keys may or may not be cadmium plated.

3.4.2 Passivation. Inserts fabricated from corrosion resistant steel, less the keys, shall be passivated in accordance with QQ-P-35.

3.5 Lubrication. Inserts with self-locking internal threads shall be solid film lubricated. The lubricant shall conform to MIL-L-46010, Type I.

3.6 Design, dimensions and tolerances. Inserts and locking keys shall be in accordance with the design dimensions and tolerances specified in MS51830, MS51831, and MS51832. Dimensions and tolerances shall apply after application of the cadmium plating or surface treatment and prior to application of the lubricant as specified in 3.5.

3.7 Construction. The insert and internal thread locking device shall be an integral, all metal unit. The keys are added and shall be provided as an integrated unit.

3.8 Threads.

3.8.1 Internal and external threads. Insert internal and external threads shall be right-hand.

3.8.2 Internal threads. Internal threads shall conform to MIL-S-8879. Sizes and dimensions shall be as specified in MS51830, MS51831 and MS51832. Thread gaging shall be performed after plating and prior to application of the lubricant, as applicable. When a locking feature is employed, the insert shall permit a bolt or screw to enter a minimum of 3/4 turn before engagement of the locking device.

3.8.3 External threads. External threads shall be in accordance with FED-STD-H28/2 except minor diameters shall be as specified in the applicable specification sheet. External threads shall be measured prior to lubrication (3.5). Acceptability of external threads shall be in accordance with FED-STD-H28/20, system 22.

3.8.3.1 Thread forming. Threads may be produced either by machining, grinding, or fully formed by a single rolling process prior to plating.

3.8.3.2 Lead threads. External lead threads shall not exceed two pitches, including chamfer.

3.8.3.3 Runout. The internal thread pitch diameter shall have a circular runout with the external pitch diameter within .006 FIM when tested as specified in 4.2.8.2.4.

3.8.4 Self locking feature. Threads in the area of the locking feature may be displaced or deformed radially in a manner which is in accordance with the applicable specification sheet. The installed insert shall provide a locking torque within the limits of table I when tested as specified in 4.2.8.3.1. (Inserts shall not be subject to locking torque tests after testing as specified in 4.2.8.3.2, 4.2.8.3.3 or 4.2.8.3.4.)

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TABLE I. Internal thread self-locking torque (inch-pounds).

Insert Internal Thread Fine or Coarse	Maximum Locking Torque	Minimum Breakaway Torque
.086	2.5	.2
.112	5	.5
.138	10	1.0
.164	15	1.5
.190	18	2.0
.250	30	3.5
.3125	60	6.5
.375	80	9.5
.4375	100	14.0
.500	150	18.0
.5625	200	24.0
.625	300	32.0
.750	400	50.0
.875	600	70.0
1.000	800	90.0

3.9 Surface texture. The surface texture of the inserts prior to plating shall be in accordance with ANSI/ASME B46.1 and shall not exceed the values stated on MS51830, MS51831 and MS51832, as applicable.

3.10 Mechanical properties. Inserts conforming to the design and dimensions of MS51830, MS51831 and MS51832 shall be capable of developing the minimum proof strength values for the tensile strength specified herein:

MS51830 - 125,000 psi
MS51831 - 160,000 psi
MS51832 - 200,000 psi

3.10.1 Axial tensile strength (proof test). The installed insert (with or without locking keys installed) shall develop the tensile strength requirements, as applicable and specified in table II when tested as specified in 4.2.8.3.2.

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TABLE II. Tensile strength (proof) requirements insert internal thread proof test.

Internal Thread Nominal Size	Tensile Stress Area in^2 1/		Tensile Load MS51830 Inserts Pounds 2/		Tensile Load MS51831 Inserts Pounds 2/		Tensile Load MS51832 Inserts Pounds 2/	
	UNJC	UNJF	UNJC	UNJF	UNJC	UNJF	UNJC	UNJF
.086	.00370	.00394	462	492				
.112	.00604	.00661	755	826				
.138	.00909	.01015	1,136	1,268				
.164	.0140	.01474	1,750	1,842				
.190	.0175	.0200	2,187	2,500	2,800	3,200	3,500	4,000
.250	.0318	.0364	3,975	4,550	5,088	5,824	6,360	7,280
.3125	.0524	.0580	6,550	7,250	8,384	9,280	10,480	11,600
.375	.0775	.0878	9,687	10,975	12,400	14,048	15,500	17,560
.4375	.1063	.1187	13,287	14,837	17,008	18,992	21,260	23,740
.500	.1419	.1599	17,737	19,987	22,704	25,584	28,380	31,980
.5625	.182	.203			29,120	32,480	36,400	40,600
.625	.226	.256			36,160	40,960	45,200	51,200
.750	.334	.373			53,440	59,680	66,800	74,600
.875	.462	.509			73,920	81,440	92,400	101,800
1.000	.606	.663			96,960	106,080	121,200	132,600

1/ Stress area per ANSI B1.1.

2/ Tensile Load = (Bolt tensile stress area, In^2) x (125, 160 or 200 KSI as applicable). It should be noted that the test loads specified in table II have been modified to reflect ANSI B1.1

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3.10.2 Resistance to pullout. The installed insert shall have a minimum resistance to pullout from the parent material as specified in table III, IV or V as applicable when tested as specified in 4.2.8.3.3

TABLE III. Pullout strength - MS51830 inserts.

Nominal Internal Thread Size	Nominal External Thread Size	Minimum Shear Engagement Area - External Thread In^2 1/ Non-Locked	Ultimate Pullout Load Lbs. 2/	Minimum Shear Engagement Area - External Thread In^2 1/ Internal Thd. Lock Part	Ultimate Pullout Load Lbs. 2/ Internal Thread Lock Part
.086	.164-32	.0157	423	.0157	423
.112	.190-32	.0302	815	.0302	815
.138	.216-28	.0329	888	.0329	888
.164	.250-28	.0669	1,806	.0669	1,806
.190	.3125-18	.1517	4,095	.0945	2,551
.250	.375-16	.2371	6,401	.1726	4,660
.3125	.4375-14	.3049	8,232	.2321	6,266
.375	.500-13	.4299	11,607	.3366	9,088
.4375	.5625-12	.5665	15,295	.4606	12,436
.500	.625-11	.7172	19,364	.5831	15,743

1/ Shear engagement area is an assembled dimensional value for the overall engaged area of the mating threaded members. It does not represent a dimension of either of the members in an unassembled condition.

2/ Pullout load = (Shear engagement area) x (27,000 psi). To compute minimum pullout load in other materials, multiply shear engagement area by applicable minimum shear strength of material.

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TABLE IV. Pullout strength - MS 51831 inserts.

Nominal Internal Thread Size	Nominal External Thread Size	Minimum Shear Engagement Area - External Thread In^2 1/ Non-Locked	Ultimate Pullout Load Lbs. 2/	Minimum Shear Engagement Area - External Thread In^2 1/ Internal Thd. Lock Part	Ultimate Pullout Load Lbs. 2/ Internal Thread Lock Part
.190	.375-16	.1901	5,132	.1156	3,121
.250	.4375-14	.2842	7,673	.1970	5,319
.3125	.500-13	.3588	9,687	.2608	7,041
.375	.5625-12	.4975	13,432	.3843	10,376
.4375	.625-11	.7172	19,364	.5831	15,743
.500	.6875-11	.8884	23,986	.7368	19,893
.5625	.8125-16	1.2493	33,731	1.0247	27,666
.625	.875-14	1.4866	40,138	1.2415	33,520
.750	1.125-12	2.4901	67,232	2.4478	66,090
.875	1.250-12	3.1370	84,699	3.0775	83,092
1.000	1.375-12	3.8381	103,628	3.7929	102,408

1/ Shear engagement area is an assembled dimensional value for the overall engaged area of the mating threaded members. It does not represent a dimension of either of the members in an unassembled condition.

2/ Pullout load = (Shear engagement area) x (27,000 psi). To compute minimum pullout load in other materials, multiply shear engagement area by applicable ultimate shear strength of material.

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TABLE V. Pullout strength - MS 51832 inserts.

Nominal Internal Thread Size	Nominal External Thread Size	Minimum Shear Engagement Area - External Thread In^2 1/ Non-Locked	Ultimate Pullout Load Lbs. 2/	Minimum Shear Engagement Area - External Thread In^2 1/ Internal Thd. Lock Part	Ultimate Pullout Load Lbs. 2/ Internal Thread Lock Part
.190	.4375-14	.2299	6,207	.1403	3,788
.250	.500-13	.2997	8,091	.2005	5,413
.3125	.5625-12	.4163	11,240	.3029	8,178
.375	.625-11	.5584	15,076	.4243	11,456
.4375	.6875-11	.8000	21,600	.6498	17,544
.500	.8125-16	1.0293	27,791	.8642	23,333
.5625	.875-14	1.3761	37,154	1.1131	30,053
.625	1.000-12	1.6420	44,334	1.2770	34,479
.750	1.250-12	2.7966	75,508	2.5505	68,863
.875	1.375-12	3.4652	93,560	3.2769	88,476
1.000	1.500-12	4.2374	114,409	4.2135	113,764

1/ Shear engagement area is an assembled dimensional value for the overall engaged area of the mating threaded members. It does not represent a dimension of either of the members in an unassembled condition.

2/ Pullout load = (Shear engagement area) x (27,000 psi). To compute minimum pullout load in other materials, multiply shear engagement area by applicable ultimate shear strength of material.

3.10.3 Torque out. The installed insert shall resist the torque values specified in table VI when installed in conformance with MS51835 and tested in accordance with 4.2.8.3.4.

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TABLE VI. Torque out.

Insert Internal Thread Size	Torque Inch-Pounds		
	MS 51830 Inserts	MS 51831 Inserts	MS 51832 Inserts
.086	10		
.112	20		
.138	30		
.164	45		
.190	72	110	180
.250	120	180	360
.3125	240	360	480
.375	320	480	600
.4375	400	600	900
.500	600	900	1,200
.5625		1,200	1,800
.625		1,800	2,400
.750		2,400	3,600
.875		3,600	4,800
1.000		4,800	6,000

3.10.4 Hardness. Inserts shall have a hardness range as specified on the applicable specification sheet (see 3.1) when tested in conformance with 4.2.8.3.8.

3.11 Metallurgical properties.

3.11.1 Discontinuities. Inserts shall not contain discontinuities which exceed the following limitations when examined in conformance with 4.2.8.3.6.

3.11.2 Cracks (see 6.3.1). Inserts shall be free from cracks in any direction or location when examined in conformance with 4.2.8.3.6.

3.11.3 Laps, seams and inclusions (see 6.3.2, 6.3.3 and 6.3.4). Inserts shall not have laps, seams and inclusions in locations as shown in figure 1 when examined in conformance with 4.2.8.3.6. Laps, seams or inclusions in other locations shall not exceed 20 percent of the thread height (sharp V thread). Care must be exercised not to confuse cracks with discontinuities, as specified herein.

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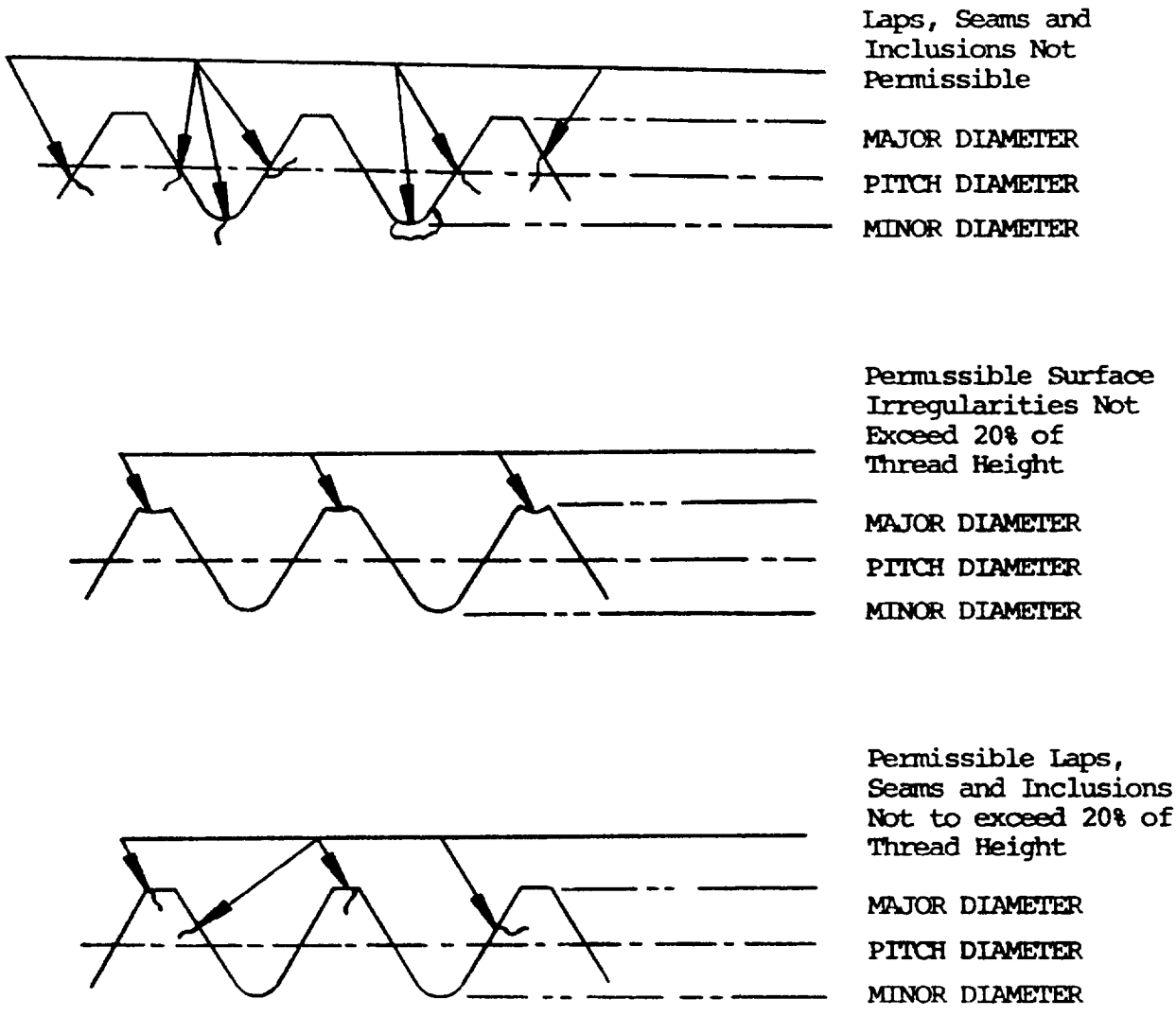


FIGURE 1. Laps, seams, inclusions and surface irregularities in thread.

3.12 Source identification mark. The insert shall be marked on the key tang or insert for internal thread sizes .190 and larger. The source identification mark for a manufacturer shall be in accordance with MIL-HDBK-57 or a private label distributor's symbol as applicable. The source identification mark shall be legible and permanent.

3.13 Workmanship. Workmanship shall be consistent with the type of product, finish, and class of thread fit specified. Inserts shall be of uniform quality and free from defects which would be detrimental to the performance of the insert.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or puchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to accept defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall be as specified in table VII. MIL-STD-105 shall be used as a guide in the development of contractors' statistical techniques to assure the inserts meet all requirements specified herein. When specified in the contract, a sampling plan shall be documented in the contractors' quality assurance plan, certified by the contractor, and subject to the approval of the Government.

4.2.1 Lot.

4.2.1.1 Inspection lot. An inspection lot shall consist of all inserts covered by a single specification sheet, of the same heat number of material and finish, nominal size and thread series, fabricated by the same production process under essentially the same conditions, and submitted for acceptance at one time. An inspection lot shall be considered as being originally part of a manufacturing lot (4.2.1.2). Any evidence of defects in the lot submitted shall constitute cause for rejection of the entire lot.

4.2.1.2 Manufacturing lot. A manufacturing lot is one continuous run of processed inserts covered by a single specification sheet of the same heat number of material and finish, nominal size and thread series, fabricated by the same production process where evidence of specified destructive type sample testing (4.2.4, 4.2.5 and 4.2.6) has been conducted and submitted for acceptance at one time. Any evidence of defects in the lot submitted shall constitute cause for rejection of the entire lot.

4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for inspection. Resubmitted lots shall be in accordance with 4.2. Such lots shall be separate from new lots, and shall be clearly identified as re-inspected lots.

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TABLE VII. Quality conformance inspection.

INSPECTION	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Group A		
Material	3.2	4.2.8.3.5
Heat Treatment	3.3	4.2.6
Cadmium Plating	3.4.1	4.2.8.2.7
Passivation	3.4.2	4.2.8.2.8
Source Identification	3.12	4.2.8.2.6
Workmanship	3.13	4.2.8.2.6
Packaging	5.1	4.2.7
Group B		
Magnetic Permeability	3.2.2	4.2.8.3.7
Threads	3.8	4.2.8.2.2, 4.2.8.2.3, 4.2.8.2.4
Surface Texture	3.9	4.2.8.2.6
Group C		
Thread Locking Feature	3.8.4	4.2.8.3.1
Mechanical Properties	3.10	4.2.4
Tensile Strength (Proof)	3.10.1	4.2.8.3.2
Resistance to Pullout	3.10.2	4.2.8.3.3
Torque Out	3.10.3	4.2.8.3.4
Hardness	3.10.4	4.2.8.3.8
Thread Discontinuities	3.11.1	4.2.8.3.6
Cracks	3.11.2	4.2.8.3.6
Laps, Seams & Inclusions	3.11.3	4.2.8.3.6

4.2.3 Sampling for visual and dimensional examination. Samples of inserts shall be taken from each lot and examined in accordance with table VIII and 4.2.

4.2.4 Sampling for tests of tensile strength, pullout and torque out. Sampling for these tests shall be in accordance with table VII, 4.2 and 4.2.1.2.

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4.2.5 Sampling for thread locking feature. Samples of self-locking inserts shall be tested in accordance with table VII, 4.2 and 4.2.1.2.

4.2.6 Samples of hardness for alloy steel inserts. Samples of alloy steel inserts shall be tested for hardness in accordance with table VII, 4.2 and 4.2.1.2.

4.2.7 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of PPP-H-1581.

TABLE VIII. Visual and dimensional examination.

DEFECT	INSPECTION METHOD
Thread size and form, not as specified. (see 3.8)	C.I.E. 1/
Locking element, not present. (see 3.8.4)	Visual
Surface finish or cadmium plating, missing or incomplete. (see 3.4)	Visual
Surface texture, nonconforming. (see 3.9)	C.I.E./Visual
Lubrication, missing or incomplete. (see 3.5)	Visual
Locking keys, missing. (see 3.6)	Visual
Thread discontinuities. (see 3.11)	C.I.E.
Runout, nonconforming. (see 3.8.3.3)	C.I.E.
Source identification mark, nonconforming. (see 3.12)	Visual
Overall length, nonconforming. (see 3.6)	C.I.E.
Dislodgeable burrs and tool marks. (see 3.13)	Visual
Chamfer on thread ends, nonconforming. (see 3.6)	Visual

1/ Commercial inspection equipment

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4.2.8 Method of inspection.

4.2.8.1 Test materials.

4.2.8.1.1 Test blocks. Test blocks shall be fabricated as specified in figure 2. Larger test blocks for multiple testing of inserts are permissible. The aluminum alloy used to prepare the test blocks shall be tested (by heat lot) to determine the ultimate shear strength. Aluminum alloys 2024 (UNS A92024) T4 and T351 tempers shall have a minimum shear strength of 37,000 psi. The 6061 (UNS A96061) T6 and T651 tempers shall have a minimum shear strength of 27,000 psi. Three coupons, per heat lot, of aluminum material shall be made and tested in conformance with MIL-STD-1312, Method 13.

4.2.8.1.2 Test specimens. Test specimens shall consist of inserts installed in conformance with MS51835 in test blocks specified in 4.2.8.1.1 (figure 2).

4.2.8.1.3 Test bolts and screws. Bolts or screws for use in all tests shall have Class 3A threads in conformance with MIL-S-8879.

4.2.8.1.3.1 Ultimate tensile strength of test bolts and screws. Bolts or screws used for testing as specified in 4.2.8.3.2, 4.2.8.3.3 and 4.2.8.3.4 shall have a minimum ultimate tensile strength of 170,000 psi for use in inserts with internal thread sizes .086 through .164 inches. Inserts with internal thread sizes .190 inches and above shall utilize bolts with a minimum ultimate tensile strength of 200,000 psi. No specific finish or material is required.

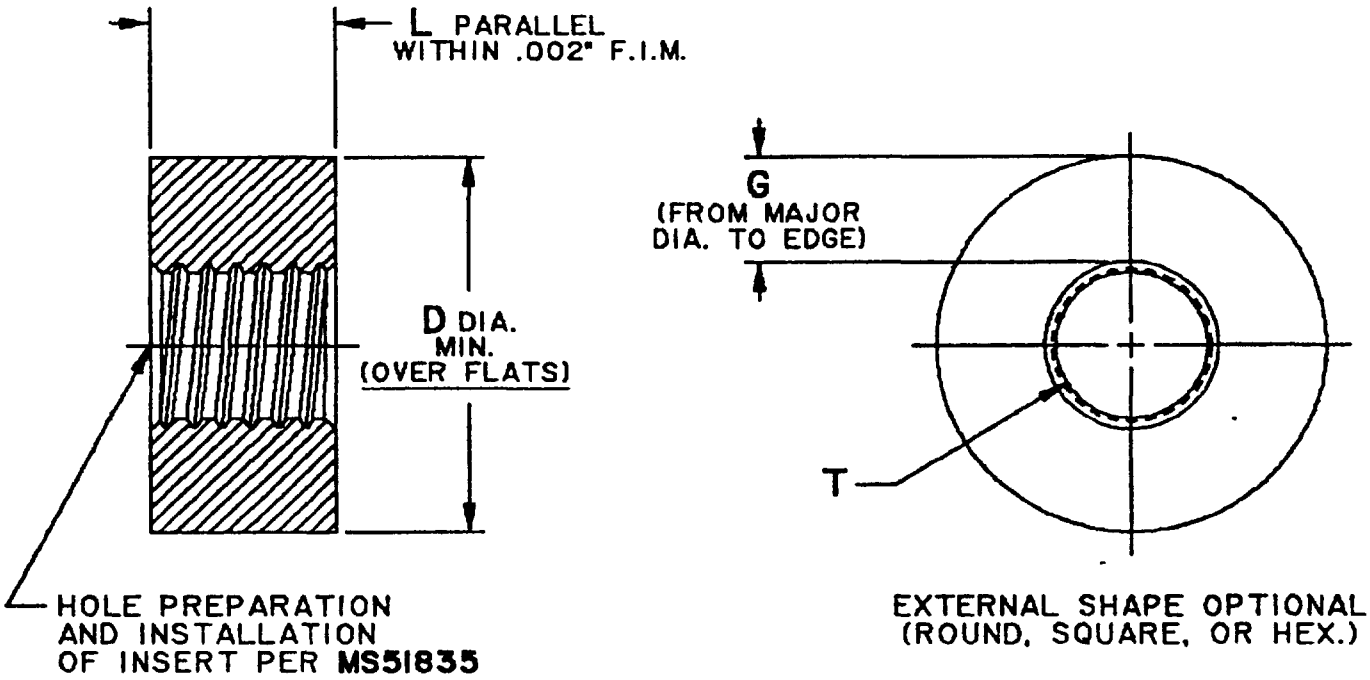
4.2.8.1.3.2 Self-locking torque test bolts and screws. Bolts or screws used in the self-locking torque test specified in 4.2.8.3.1.1 shall be made of alloy or carbon steel heat treated to 125,000 psi minimum ultimate tensile strength and cadmium plated per QQ-P-416, Type 1 or Type II, Class 2 or 3 when testing alloy steel inserts. Corrosion resistant steel bolts with a minimum tensile strength of 125,000 psi shall be used when testing corrosion resistant steel inserts. Test bolts shall be selected within the lower 50 percent of Class 3A thread tolerance range of the pitch diameter. Plating or lubrication may be used on test bolts to prevent galling.

4.2.8.2 Examination methods.

4.2.8.2.1 Visual and dimensional examination. Samples of the inserts shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with table VIII and the following.

4.2.8.2.2 External threads. Prior to assembly of the keys, samples of the inserts' external threads shall be inspected for thread form in conformance with the gaging requirements of FED-STD-H28/20, System 22. The minor diameter dimensions shall be per the applicable specification sheet (see 3.1). Assembled inserts (after key installation) shall be checked in a tapped hole gage in accordance with MS51835.

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1. DIMENSIONS:

   T = NOMINAL EXTERNAL THREAD DIAMETER OF APPLICABLE INSERT.
   L = LENGTH OF APPLICABLE INSERT PLUS .063" MINIMUM.
   G = TO BE A MINIMUM OF 1/2 T
   D = TO BE 1.00", 2.00", 3.00" OR LARGER AS REQUIRED.

2. MATERIAL:

   ALUMINUM ALLOY, 2024-T4 OR 2024-T351 PER QQ-A-225/6 OR QQ-A-250/4 (3.8.4 AND 3.10.1).

   ALUMINUM ALLOY, 6061-T6 OR 6061-T651 PER QQ-A-225/8 OR QQ-A-250/11 (3.8.4, 3.10.2 AND 3 10.3).

FIGURE 2. Test coupon.

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4.2.8.2.3 Internal threads. Internal threads shall be inspected in accordance with MIL-S-8879.

4.2.8.2.4 Runout. The runout of the internal and external thread pitch diameters shall be checked following examination per 4.2.8.2.2 and 4.2.8.2.3. The internal thread pitch diameter shall be inspected for runout with the external thread pitch diameter using a comparator. Runout shall be checked while rotating the insert on a fixed internal thread plug gage and observing the circular runout of the external thread pitch line or minor diameter line relative to the thread form.

4.2.8.2.5 Lubrication. The solid film lubricant coating shall be examined visually and microscopically at a magnification of 10X for uniformity in color, smoothness, and thickness, and evidence of cracks, bubbles, blisters, runs, foreign matter, grit separation of ingredients, and any other surface imperfections.

4.2.8.2.6 Finish, dimensions, source identification, workmanship and surface texture inspection. Finish, dimensions, source identification, workmanship and surface texture shall be inspected visually to determine conformance with the requirements of 3.4 and 3.6.

4.2.8.2.7 Cadmium plating. Cadmium plating shall be inspected for conformance with the requirements of 3.4.1.

4.2.8.2.8 Passivation. Passivation shall be inspected for conformance with the requirements of 3.4.2.

4.2.8.3 Tests methods.

4.2.8.3.1 Internal thread self-locking test. Test specimens installed in test blocks as specified in 4.2.8.1.2 (figure 2) shall be used for torque testing of the internal locking feature when applicable. Other methods that capture the insert are acceptable provided that supporting test data is available to show equivalency.

4.2.8.3.1.1 Screw-locking torque. The locking torque test specified in 3.8.4 shall consist of a 15 cycle room temperature torque test, using test bolts or screws conforming to 4.2.8.1.3.2 with sufficient thread length to extend beyond the locking feature a minimum of two (2) pitches. A new bolt or screw and a new test specimen shall be used for each complete 15 cycle test. Bolts and screws shall assemble freely, with the fingers, up to the locking feature. The bolt or screw shall be engaged or disengaged from the assembled insert self-locking area for 15 full installation and removal cycles with no axial load on the insert. The test shall be run at a rate slow enough to yield a dependable measure of torque and avoid heating of the bolt. A bolt shall be considered fully installed when two threads extend past the end of the locking feature of the insert; the removal cycle shall be considered complete when the locking feature is disengaged. The maximum locking torque encountered on any installation or removal cycle shall not exceed the values specified in table I. The minimum breakaway torque for any cycle shall not be less than the values specified in table I to meet the requirements of 3.8.4.

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4.2.8.3.2 Axial tensile strength (proof) test. Test specimens installed into test blocks (figure 2) as specified in 4.2.8.1.2 shall be used for axial strength testing. The test bolt shall assemble freely into the insert prior to engaging the self-locking device with finger torque and shall be of sufficient length to extend through the insert a minimum of two (2) pitches. The test shall be conducted in fixtures as shown in figure 3. The minimum axial load specified in table II, as applicable, shall then be applied to the assembly and failure of the insert shall not occur below those loads specified in table II. In the event of bolt failure below the applicable axial strength rating, the test shall be repeated until the applicable axial strength rating of the insert is exceeded. Rate of loading shall not exceed 100,000 psi per minute based on the shank area of the bolt.

4.2.8.3.3 Resistance to pullout test. The test shall be conducted in fixtures as shown in figure 3. Test specimens installed into test blocks (figure 2) as specified in 4.2.8.1.2 shall be used for resistance to pullout tests. The procedure used shall be the same as required for 4.2.8.3.2 except the minimum resistance to pullout strength specified in tables III, IV or V, as applicable, shall be applied to the assembly. Failure of the insert to stay in the test block shall not occur below the loads specified in tables III, IV or V. In the event of bolt failure below the applicable pullout strength rating, the test shall be considered acceptable provided that minimum ultimate tensile strength of the bolt used was achieved. Rate of loading shall not exceed 100,000 psi per minute based on the shank area of the bolt.

4.2.8.3.4 Torque out test. Test specimens installed per 4.2.8.1.2 in test blocks (figure 2) shall be used for torque out tests. The torque values listed in table VI, with no axial load exerted on the test coupon shall be reached, and the bolt capable of being removed from the insert before the test is considered successful. The insert must remain in the test coupon during the test and after bolt removal. Any method shown in figure 4 may be used. Should the bolt fail or the spacer collapse prior to reaching the torque values shown, the test shall be considered successful.

4.2.8.3.5 Raw material chemical analysis. Raw material chemical analysis certification may be accepted in lieu of test data.

4.2.8.3.6 Discontinuities. When visual inspection discloses discontinuities and shows cause for further examination, magnetic particle inspection shall be performed in accordance with MIL-STD-1949 for alloy steel, and penetrant inspection in accordance with MIL-STD-6866 for corrosion-resistant steel.

4.2.8.3.7 Magnetic permeability. Type A286 corrosion-resistant steel inserts shall be tested to determine magnetic permeability in accordance with MIL-I-17214 to meet the requirements of 3.2.2.

4.2.8.3.8 Hardness test. Alloy steel inserts shall be tested for hardness in accordance with MIL-STD-1312, Test 6 to determine compliance with 3.10.4.

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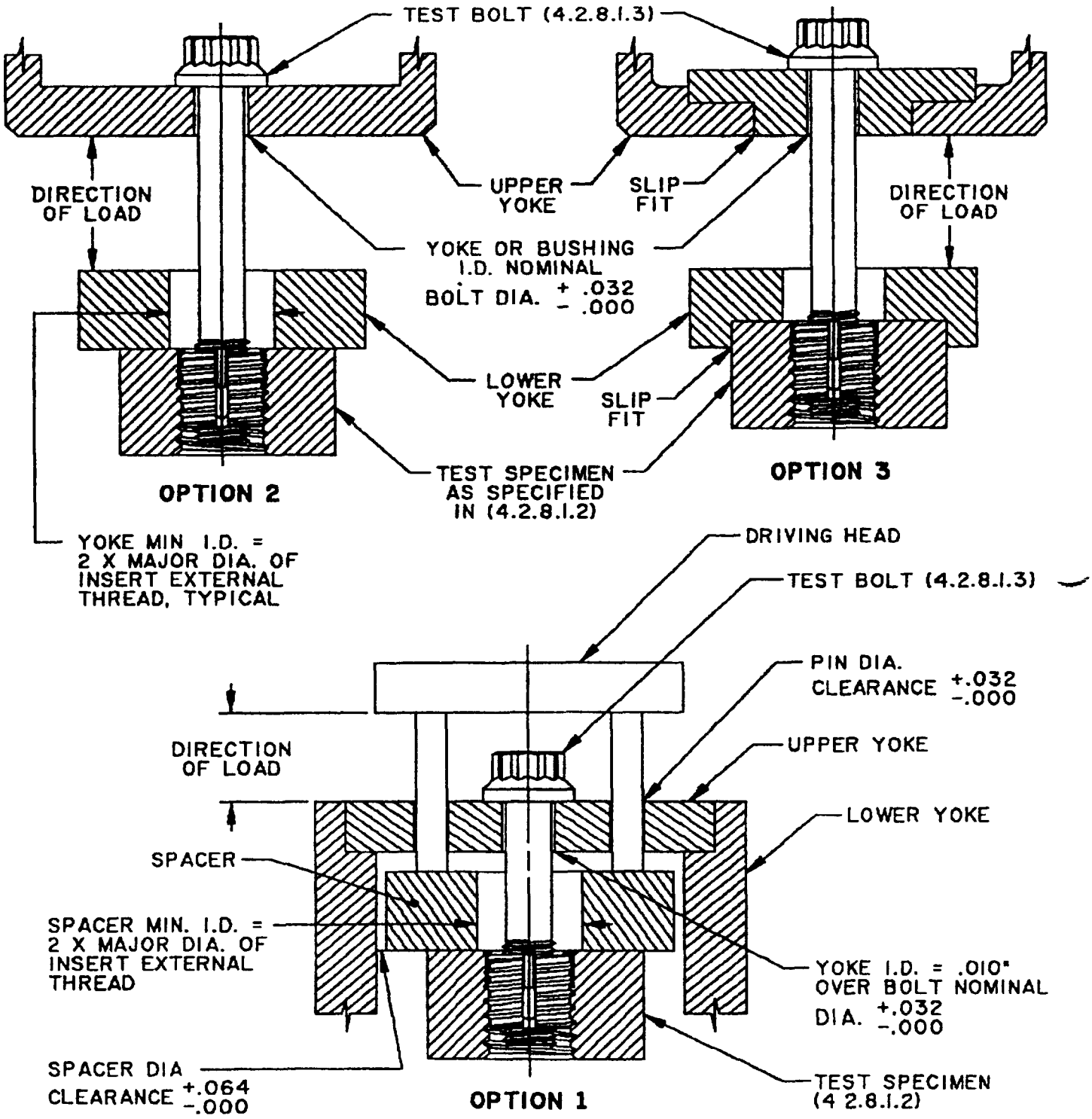


FIGURE 3. Tensile strength (3.10.1) and resistance to pull-out (3.10.2) test set-up(s).

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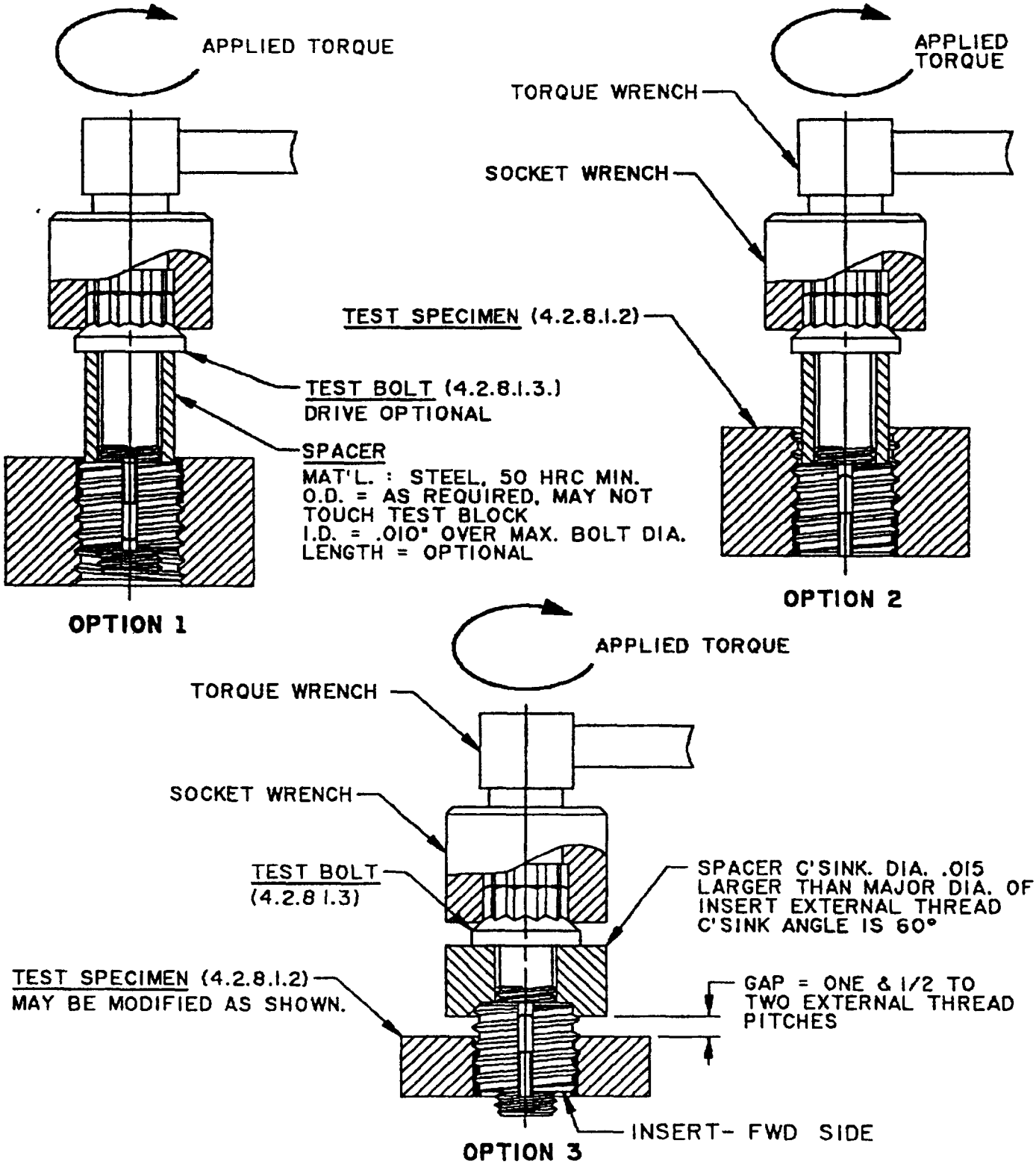


FIGURE 4. Torque-out test set-up(s) (3.8.4).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2e).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Inserts covered by this specification are intended as a general purpose fastener with depth locating keys and a mechanical lock to resist rotation when installed.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Title, number and date of applicable specification sheet(see 3.1).
(c) Military part number.
(d) Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents references (see 2.1).
(e) Level (degree) of protection, in accordance with PPP-H-1581, ordering data (see 5.1).

6.3 Definitions.

6.3.1 Crack. A crack is defined as a clean crystalline break passing through the grain or grain boundary without the inclusion of foreign elements.

6.3.2 Lap. A lap is a surface defect appearing as a seam, caused by the folding over of metal fins or sharp corners and then rolling or forging them into the surface, but not welding them.

6.3.3 Seam. A seam is an unwelded fold or lap which appears as an opening in the raw material as received from the source.

6.3.4 Inclusions. Inclusions are non-metallic materials in a solid metallic matrix.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Keyword Listing.

Screw Thread Insert
Locked-In Insert
Key Locked Insert
Locking Keys

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Navy - AS
Air Force - 99

Preparing activity:
Army - AR

(Project No. 5340-2027)

Review activities:
Army - AV, ER, MI
Navy - SH
Air Force - 82
DLA - IS

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3 DOCUMENT TITLE
INSERT, SCREW THREAD - LOCKED IN, KEY LOCKED, GENERAL SPECIFICATION FOR

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets if needed)*

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